Haoyang Li 李昊阳

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Education

University of Virginia — Charlottesville, VA, USA
Ph.D. Student in Systems Engineering — 2026–2031 *(Anticipated)*

- **Advisor:** Prof. Hongru Du
- **Research Interest:** AI for Science, Large Language Model Agents, Complex Systems, Computational Social Science, Social Stratification, Social Networks

Hong Kong Baptist University — Hong Kong SAR, China
B.S.S.C. (Hons.) in Sociology, Minor in Computer Science — 2022–2026

- **Classification:** First Class Honors
- **GPA:** cGPA: 3.63/4.0, Major: 3.70/4.0, Minor: 3.52/4.0
- **Relevant Coursework:** Intro to AI & ML (A-) | NLP & LLM (A) | Social Stat. (A) | Intro to Python & its Applications (A-) | Problem Solving Using OOP (A) | Calc., Prob. & Stat. for CS (A-) | Sociological Classical Theory (A) | Sociological Theory II (A-) | Social Networks (A-)

Publications

* Equal contribution
✉ Corresponding

1. [Under Review] **Haoyang Li**, Runzhou Liu, Yao Li, Amy Wesolowski, Sen Pei, Hongru Du✉. (2026). Theory-informed generative agents for modeling human mobility. *Preprint available at Research Square*.
2. [Under Review] Runzhou Liu, Claire Jong, **Haoyang Li**, Yiming Cao, Qing Yao, Teresa Yamana, Sen Pei✉, Hongru Du✉. (2026). Simulating population compliance with pandemic interventions using Large Language Models. *medRxiv 2026.05.12.26352942*.
3. [Under Review] Xiao Liu*, **Haoyang Li***, Songwei Li, Hongbo Fang, Fengli Xu, Feng Shi, & James Evans. (2026). Markets, Not Planners: Decentralized Orchestration of LLM Agents with Private Information. *arXiv preprint arXiv:2608.23867*.
4. [Under Review] Qingbin Zeng, Ruotong Zhao, Jinzhu Mao, **Haoyang Li**, Fengli Xu✉, & Yong Li. (2025). CrimeMind: Simulating urban crime with multi-modal LLM agents. *arXiv preprint arXiv:2506.05981*.
5. [Working Paper] **Haoyang Li**, Xiao Jia, & Zhanzhan Zhao✉. (2025). The emergence of altruism in Large-Language-Model agents society. *arXiv preprint arXiv:2509.22537*.

Research Experience

- **Graduate Research Assistant** | Du Lab, **University of Virginia** — 2026.08 – Present
- **Research Intern** — 2025.07 – 2026.07
  - Supervised by **Prof. Hongru Du**, Assistant Professor, Department of Systems and Information Engineering, University of Virginia
  - *Research Focus*: AI for Science, Modeling Human Behavior through LLM Agents

Research Intern | Knowledge Lab, **University of Chicago** — 2025.10 – 2026.08

- Supervised by **Prof. James Evans**, Max Palevsky Professor, Department of Sociology, University of Chicago
- *Research Focus*: AI Scientists for Market Allocation

Research Intern | Future Intelligence Lab, **Tsinghua University** — 2025.03 – 2026.08

- Supervised by **Prof. Fengli Xu**, Assistant Professor, Department of Electronic Engineering, Tsinghua University
- *Research Focus*: LLM Agents, Social Simulation

Research Experience (Cont'd)

Research Assistant | School of Artificial Intelligence, **The Chinese University of Hong Kong, Shenzhen** 2025.07 – 2025.11

- Supervised by **Prof. Zhanzhan Zhao**, Assistant Professor, School of Artificial Intelligence, The Chinese University of Hong Kong, Shenzhen
- *Research Focus*: LLM and Game Theory

Research Assistant | Department of Sociology, **Hong Kong Baptist University** 2024.03 – 2024.12

- Supervised by **Prof. Yinni Peng**, Professor, Department of Sociology, Hong Kong Baptist University

Research Assistant | Department of Sociology, **Hong Kong Baptist University** 2024.03 – 2024.06

- Supervised by **Prof. Jin Jiang**, Associate Professor, Department of Sociology, Hong Kong Baptist University

Awards and Honors

- **Halloran Scholarship**, 18th SISMID, Emory University (Full Reg. Waiver) 2026.03
- **President's Honor Roll**, Hong Kong Baptist University Semester 1, 2022–2023; Semester 1&2, 2023–2024; Semester 1&2, 2024–2025; Semester 1, 2025–2026
- **Dean's List**, Hong Kong Baptist University Semester 2, 2022–2023; Semester 2, 2025–2026
- **HKSAR Government Talent Development Scholarship**, HK$10,000 2023.04

Talks & Posters

- **Invited Session Presentation**, INFORMS 2026 Annual Meeting 2026.11
  Title: Theory-Informed Generative Agents for Modeling Human Mobility • **Venue**: San Francisco, CA, USA • [Slides] • [Certificate]
- **Mainland Undergraduate Admissions Talk**, Hong Kong Baptist University 2026.05
- **Poster Presentation**, ICAIS 2025 2025.11
  Title: Dispersed Knowledge, Decentralized Decisions: A Labor Market for Multi-Agent Task Allocation • **Venue**: The 1st International Conference on AI Scientists (ICAIS 2025), Beijing, China • [ICAIS Website] • [Poster]

Teaching

- **Teaching Assistant**, Hong Kong Baptist University 2024.01 – 2024.04
  Course: COMP1005 Essence of Computing • [Course Outline]

Skills

- **Languages**: Chinese Mandarin (Native), Chinese Cantonese (Listening), English (Working Language), LaTeX
- **Programming & Software**: `Python`, `Java`, `C`, `SQLite`, `MATLAB`, `Stata`, `SPSS`, `Microsoft Office`
- **Interests**:
  - **Chinese Writing**: won the *First Prize* of the 24th National New Concept Writing Contest
  - **Hiking**: I left many footprints along the *MacLehose Trail* and other mountains across Hong Kong and Shenzhen